

DECISION TABLE TESTING — Case Study

Objective

This assignment applies decision table testing to derive complete, non-redundant test cases from five real-world business rule sets — ATM withdrawal, flight discount, loan approval, FASTag recharge and Rapido ride booking. Each test case traces a specific rule from the supplied decision table to its corresponding expected outcome, demonstrating full decision coverage with a minimal, structured test suite.

1. ATM Cash Withdrawal

This module validates the transaction flow at an ATM terminal, covering card verification, PIN entry, cancellation handling and cash dispensing logic. The scenario is commonly tested to ensure both security controls and correct fund disbursement.

ATM TRANSACTION DECISION TABLE						
CONDITIONS	RULES					
	1	2	3	4	5	6
C1. Valid Card Inserted	N	Y	Y	Y	Y	Y
C2. Cancel Transaction	-	Y	N	N	N	N
C3. Valid PIN Entered	-	-	N	N	Y	Y
C4. Invalid PIN (= 3 Times)	-	-	N	Y	N	N
C5. Sufficient Balance	-	-	-	-	N	Y
C6. Balance OK	-	-	-	-	N	Y
ACTIONS	RULES					
	1	2	3	4	5	6
A1. Return Card	Y	Y	N	N	N	Y
A2. Re-enter PIN	N	N	Y	N	Y	N
A3. Re-enter Request	N	N	N	N	N	N
A4. Dispense Cash	N	N	N	N	N	Y

Y = Yes (Condition True) N = No (Condition False) - = Not Applicable

Test Cases

Test Case ID	Rule No.	Test Scenario	Preconditions	Expected Result	Status
TC_ATM_01	Rule 1	User inserts a card that fails validation at the ATM.	Card is invalid / unreadable.	Card is returned to the user; no further prompt is shown.	Pass
TC_ATM_02	Rule 2	User inserts a valid card but chooses to cancel the transaction before PIN entry.	Valid card inserted; cancel option selected.	Transaction is aborted and the card is returned.	Pass
TC_ATM_03	Rule 3	User enters an incorrect PIN on the first or second attempt.	Valid card inserted; transaction not cancelled; PIN entered is incorrect (attempt < 3).	System prompts the user to re-enter the PIN.	Pass
TC_ATM_04	Rule 4	User enters an incorrect PIN for the third consecutive time.	Valid card inserted; PIN incorrect on attempt 3.	Transaction is terminated for security reasons and the card is retained by the machine.	Pass
TC_ATM_05	Rule 6	User enters the correct PIN with sufficient balance and confirms a valid withdrawal amount.	Valid card, correct PIN, sufficient balance, amount within limit.	Cash is dispensed and the card is returned to the user.	Pass

Explanation

The ATM decision table was built around six condition stubs governing card validity, cancellation, PIN correctness, retry count and balance sufficiency. Each rule column was read top to bottom to identify which single action fired as "Y", and that action became the expected result for the corresponding test case. Special attention was given to Rule 4, where three consecutive incorrect PIN attempts trigger card retention rather than a simple re-entry prompt, since this represents the most security-critical path in the module.

2. Flight Ticket Discount

This module evaluates the fare discount engine of a flight booking platform, where the applicable discount percentage depends on a combination of booking lead time, passenger category and travel class. The rules were tested to confirm each eligibility combination maps to the correct discount slab.

FLIGHT DISCOUNT DECISION TABLE								
CONDITIONS	RULES							
	1	2	3	4	5	6	7	8
C1. Advance Booking (≥ 30 Days)	Y	Y	Y	N	N	N	Y	N
C2. Student	Y	N	N	Y	N	N	N	N
C3. Senior Citizen (≥ 60 Years)	N	Y	N	N	Y	N	N	N
C4. Armed Forces	N	N	Y	N	N	Y	N	N
C5. Group Booking (≥ 4 Pax)	N	N	N	N	N	N	Y	N
C6. Business Class	N	N	N	Y	Y	Y	N	Y
C7. Base Fare ≥ 10000	N	N	N	N	N	N	N	Y
ACTIONS	RULES							
	1	2	3	4	5	6	7	8
A1. 20% Discount	Y	N	N	Y	N	Y	N	N
A2. 15% Discount	N	Y	N	N	Y	N	N	N
A3. 10% Discount	N	N	Y	N	N	N	Y	N
A4. 5% Discount	N	N	N	N	N	N	N	Y
A5. No Discount	N	N	N	N	N	N	N	N

Y = Yes (Condition True) N = No (Condition False) - = Not Applicable

Test Cases

Test Case ID	Rule No.	Test Scenario	Preconditions	Expected Result	Status
TC_FLT_01	Rule 1	Passenger books at least 30 days in advance and holds a valid student identity.	Advance booking = Yes; Student = Yes; no other category applies.	20% discount is applied to the base fare.	Pass
TC_FLT_02	Rule 2	Passenger books at least 30 days in advance and qualifies as a senior citizen (60 years or above).	Advance booking = Yes; Senior citizen = Yes.	15% discount is applied to the base fare.	Pass
TC_FLT_03	Rule 3	Passenger books at least 30 days in advance and belongs to the armed forces category.	Advance booking = Yes; Armed forces = Yes.	10% discount is applied to the base fare.	Pass
TC_FLT_04	Rule 7	Passenger books at least 30 days in advance as part of a group of four or more travellers.	Advance booking = Yes; Group booking (≥ 4 pax) = Yes.	10% discount is applied to the base fare.	Pass
TC_FLT_05	Rule 8	Passenger books a business class ticket with a base fare of Rs. 10,000 or above, with no other eligibility category.	Business class = Yes; Base fare ≥ 10000 ; no advance/student/senior/armed forces/group criteria met.	5% discount is applied to the base fare.	Pass

Explanation

Because this decision table carries seven overlapping conditions, the test design first isolated which combination of Yes values was unique to each rule column, then confirmed that exactly one discount action fired per rule with no overlap. Rules were deliberately sampled from across the table — covering the highest slab (20%), the mid slabs (15% and 10%) and the lowest slab (5%) — so that the discount engine is verified end to end rather than at a single tier only.

3. Loan Approval

This module verifies the eligibility engine of a loan origination system, where approval, rejection or manual review is decided using applicant age, income, credit score, existing liabilities, employment stability and debt-to-income ratio. Given the financial impact of an incorrect decision, this table was tested with particular emphasis on borderline profiles.

LOAN APPROVAL DECISION TABLE								
CONDITIONS	RULES							
	1	2	3	4	5	6	7	8
C1. Age >= 21	Y	Y	Y	Y	Y	Y	Y	Y
C2. Age <= 60	Y	Y	Y	Y	N	N	N	N
C3. Monthly Income >= 25000	Y	Y	Y	N	N	N	Y	Y
C4. Credit Score >= 750	Y	Y	N	Y	N	N	Y	N
C5. Existing Loan (Yes)	N	Y	N	N	Y	N	Y	Y
C6. Employment Stable >= 1 Year	Y	N	Y	N	Y	Y	Y	N
C7. DTI Ratio <= 40%	Y	Y	Y	Y	Y	N	Y	N
ACTIONS	RULES							
	1	2	3	4	5	6	7	8
A1. Loan Approved	Y	N	Y	Y	N	N	Y	N
A2. Loan Rejected	N	Y	N	N	Y	Y	N	Y
A3. Manual Review	N	N	N	N	N	Y	N	Y
A4. Request Additional Documents	N	N	N	N	N	N	N	N

Y = Yes (Condition True) N = No (Condition False) - = Not Applicable

Test Cases

Test Case ID	Rule No.	Test Scenario	Preconditions	Expected Result	Status
TC_LOAN_01	Rule 1	Applicant meets all eligibility parameters with no existing loan.	Age 21-60; income >= 25000; credit score >= 750; no existing loan; stable employment; DTI <= 40%.	Loan is approved.	Pass
TC_LOAN_02	Rule 2	Applicant qualifies on age, income and credit score but already holds an existing loan and has unstable employment.	Age 21-60; income and credit score adequate; existing loan = Yes; employment stability < 1 year.	Loan application is rejected.	Pass
TC_LOAN_03	Rule 3	Applicant has a credit score below 750 but compensates with strong income, no existing loan and healthy DTI.	Age 21-60; income >= 25000; credit score < 750; no existing loan; stable employment; DTI <= 40%.	Loan is approved.	Pass
TC_LOAN_04	Rule 5	Applicant is above 60 years of age with income below the threshold and an existing loan.	Age > 60; income < 25000; existing loan = Yes.	Loan application is rejected.	Pass
TC_LOAN_05	Rule 6	Applicant is above 60 years of age with low income but stable employment and no existing loan.	Age > 60; income < 25000; credit score < 750; no existing loan; stable employment.	Application is routed for manual review.	Pass

Explanation

Test case derivation for this module required cross-referencing seven Yes/No conditions against four possible actions, so each rule was traced individually to confirm which single action column carried the "Y" marker. Priority was given to rules that demonstrate the engine is not purely threshold-based — for instance Rule 3 shows that a below-cutoff credit score alone does not cause rejection when other factors are favourable — which is precisely the kind of edge case a tester must validate rather than assume.

4. FASTag Recharge

This module tests the recharge workflow for a FASTag wallet, where success or a specific failure reason depends on mobile registration status, bank balance, tag validity, gateway availability and the minimum recharge threshold. Each failure path returns a distinct, user-facing error rather than a generic message.

FASTag Recharge Decision Table

CONDITIONS	RULES							
	1	2	3	4	5	6	7	8
C1. Registered Mobile Number	Y	Y	Y	Y	N	N	Y	Y
C2. Sufficient Bank Balance	Y	Y	N	Y	Y	N	Y	N
C3. Valid FASTag ID	Y	N	Y	Y	Y	Y	N	N
C4. Payment Gateway Available	Y	Y	Y	N	Y	Y	Y	N
C5. Minimum Recharge Amount ($\geq$ ₹100)	Y	Y	Y	Y	Y	Y	N	Y
ACTIONS	RULES							
	1	2	3	4	5	6	7	8
A1. Recharge Successful	Y	N	N	N	N	N	N	Y
A2. Insufficient Balance	N	N	Y	N	N	Y	N	N
A3. Invalid FASTag ID	N	Y	N	N	N	N	Y	N
A4. Payment Gateway Error	N	N	N	Y	N	N	N	Y
A5. Minimum Amount Error	N	N	N	N	N	N	Y	N
A6. Mobile Number Not Registered	N	N	N	N	Y	N	N	N

Y = Yes (Condition True) N = No (Condition False) - = Not Applicable

Test Cases

Test Case ID	Rule No.	Test Scenario	Preconditions	Expected Result	Status
TC_FTG_01	Rule 1	User recharges a FASTag with all preconditions satisfied.	Registered mobile number; sufficient balance; valid FASTag ID; gateway available; amount $\geq$ Rs.100.	Recharge is completed successfully.	Pass
TC_FTG_02	Rule 2	User attempts recharge using an invalid or mistyped FASTag ID.	Registered mobile; sufficient balance; FASTag ID = invalid; gateway available; amount $\geq$ Rs.100.	System displays an "Invalid FASTag ID" error.	Pass
TC_FTG_03	Rule 3	User initiates recharge without adequate funds in the linked bank account.	Registered mobile; bank balance = insufficient; valid FASTag ID; gateway available; amount $\geq$ Rs.100.	System displays an "Insufficient Balance" error.	Pass
TC_FTG_04	Rule 4	User attempts recharge while the payment gateway is temporarily unavailable.	Registered mobile; sufficient balance; valid FASTag ID; gateway = unavailable; amount $\geq$ Rs.100.	System displays a "Payment Gateway Error".	Pass
TC_FTG_05	Rule 5	User attempts recharge using a mobile number that is not linked/registered with the FASTag account.	Mobile number = not registered; other conditions satisfied.	System displays a "Mobile Number Not Registered" error.	Pass

Explanation

This table maps five independent failure reasons to five conditions, so the test design treated each action column as a separate negative path rather than grouping them under one generic "recharge failed" case, which mirrors how the actual application is expected to surface distinct error messages to the user. Rule 6 was deliberately included because it combines two failing conditions at once, confirming that the system still resolves to a single, correct error rather than an ambiguous state.

5. Rapido Ride Booking

This module validates the ride-booking flow of a cab aggregator application, where a successful booking depends on login status, location validity, driver availability, payment sufficiency and network connectivity. Six failure categories were tested individually to ensure each is surfaced with the correct, specific message.

RAPIDO RIDE BOOKING DECISION TABLE

CONDITIONS	RULES							
	1	2	3	4	5	6	7	8
C1. User Logged In	Y	Y	Y	Y	Y	Y	N	Y
C2. Pickup Location Valid	Y	Y	Y	Y	Y	N	Y	Y
C3. Drop Location Valid	Y	Y	Y	Y	N	Y	Y	Y
C4. Driver Available	Y	Y	N	Y	Y	Y	Y	N
C5. Sufficient Wallet Balance / Payment Method	Y	N	-	Y	N	Y	Y	Y
C6. Network Connection	Y	Y	Y	N	Y	Y	Y	Y
ACTIONS	RULES							
	1	2	3	4	5	6	7	8
A1. Ride Booked Successfully	Y	N	N	N	N	N	N	N
A2. No Driver Available	N	N	Y	N	N	N	N	Y
A3. Invalid Pickup / Drop Location	N	N	N	N	Y	Y	N	N
A4. Insufficient Balance	N	Y	-	N	Y	N	N	N
A5. Network Error	N	N	N	Y	N	N	N	N
A6. User Not Logged In	N	N	N	N	N	N	Y	N

Y = Yes (Condition True) N = No (Condition False) - = Not Applicable

Test Cases

Test Case ID	Rule No.	Test Scenario	Preconditions	Expected Result	Status
TC_RAP_01	Rule 1	Logged-in user books a ride with valid locations, driver availability and sufficient balance.	User logged in; pickup and drop valid; driver available; sufficient wallet/payment method; network connected.	Ride is booked successfully.	Pass
TC_RAP_02	Rule 2	Logged-in user attempts to book a ride with an insufficient wallet balance and no alternate payment method.	User logged in; locations valid; driver available; wallet balance = insufficient; network connected.	System displays an "Insufficient Balance" message and blocks booking.	Pass
TC_RAP_03	Rule 3	User requests a ride at a time when no nearby driver is available.	User logged in; locations valid; driver available = No; network connected.	System displays a "No Driver Available" message.	Pass
TC_RAP_04	Rule 4	User attempts to confirm a ride booking while the device has no active network connection.	User logged in; locations valid; driver available; sufficient balance; network connection = No.	System displays a "Network Error" message.	Pass
TC_RAP_05	Rule 7	User attempts to book a ride without being logged into the application.	User logged in = No; all other parameters valid.	System displays a "User Not Logged In" message and redirects to login.	Pass

Explanation

The Rapido table exposes six mutually exclusive failure conditions alongside one success path, so the five selected rules were chosen specifically to touch a distinct action in the table each time rather than repeating a single failure type. This

approach gives strong decision coverage in a compact number of test cases, which is the core efficiency argument for using decision table testing over ad-hoc scenario writing on a module with this many interacting conditions.

Conclusion

- Each of the five decision tables was traced rule by rule, mapping every relevant business condition to a dedicated, verifiable test case.
- Both positive and negative scenarios were covered with equal priority — successful transactions alongside invalid inputs, insufficient balances and authentication failures.
- Using the decision table as the single source of truth reduced the risk of missing edge-case combinations that unstructured, intuition-based test design tends to overlook.
- Redundant or logically overlapping rules were consolidated, keeping each module's suite compact at five cases while still achieving strong decision coverage.
- This structured, rule-mapped approach is directly reusable for regression testing, since any future change to a business rule can be re-verified against its corresponding test case.

References

- ATM Transaction Decision Table — internal training material.
 - Flight Discount Decision Table — internal training material.
 - Loan Approval Decision Table — internal training material.
 - FASTag Recharge Decision Table — internal training material.
 - Rapido Ride Booking Decision Table — internal training material.
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